

Full Stack Resource Estimation of Titanium Dioxide for Photocatalytic Wastewater Treatment

Hisham Kazim

Department of Physics, University of Wisconsin-Madison, Madison, United States of America

Abstract

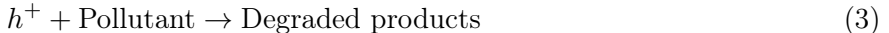
Photocatalytic wastewater treatment with TiO_2 has strong correlations between the Ti 3d and O 2p orbitals that govern electron-hole separation, pivotal in the photocatalytic process. Density functional theory is useful to model such systems, but suffers from strong electron correlations, and the classical full configuration interaction scales combinatorially in the active space. This computational study estimates the full-stack, fault-tolerant quantum resources required to compute the energy of a Ti_2O_4 anatase fragment within a chemical accuracy of 1.6 mHa. Three block encodings are compared: the Pauli linear combination of unitaries (PauliLCU), double factorization (DF), and tensor hypercontraction (THC), across six active spaces from $(4e, 4o)$ to $(24e, 24o)$. Full-configuration interaction on the $(12e, 12o)$ target verifies that the correlation energy missed by Hartree-Fock (39.7 mHa) exceeds the QPE precision, thereby validating the analysis. The results show that tensor hypercontraction reduces the T-gate count by three orders of magnitude relative to the Pauli encoding. Pushing this result to a 50-orbital target, tensor hypercontraction predicts roughly 4×10^{10} T-gates and 2.4×10^5 physical qubits at a physical error rate of 10^{-3} , which is comparable to the resources reported for the FeMoco active site even though the chemical system is vastly different. These results show the possibility of utilizing these resources in the era of fault-tolerant quantum computing. Additional considerations, such as the pollutant-photocatalyst interaction and more realistic geometries, should help provide more realistic estimates and validate the current outcomes.

Keywords: *photocatalysts, tensor hypercontraction, double factorization, wastewater treatment, resource estimation.*

1 Problem Statement and Motivation

Emerging contaminants have become a major issue in waterway systems with more than 500 million tons annually, resulting in more than 1.7 million deaths annually [1, 2]. Emerging contaminants include dyes, pesticides, antibiotics, personal care products, and heavy metals [3]. There are diverse analytical methods for remediating these emerging contaminants, including adsorption, membrane processes, advanced oxidation, electrocoagulation, and photocatalysis [4–7]. Photocatalysis uses a catalyst to break down pollutants when exposed to light. A photocatalyst is applied to remediate and reduce contaminants. The process is based on the formation of an electron-hole pair as the electron is excited to a higher energy level. The photocatalyst is illuminated as the light’s wavelength exceeds the bandgap, causing the transition to occur where a hole is created instead of an electron [8]. Equations (1)–(3) represent the process of photocatalysis. Equation (1) shows that the photocatalyst absorbs light energy $h\nu$, using TiO_2 as the photocatalyst, generating an electron e^- and a positive hole h^+ . As the electron and hole recombine, heat is released. Equation (2) shows that the presence

of the electron-hole pair at the surface of the catalyst enables redox reactions to occur, serving as the mechanism for photocatalytic treatment. Equation (3) shows how the hole in the valence band behaves in the reaction. This reaction breaks down the target pollutant and its effect. Specific photocatalysts are chosen depending on the chemical makeup of the contaminant, as the high oxidation capacity can destroy the contaminant structure [9]. The most common photocatalyst is TiO_2 due to its stability, minimal toxicity, and other chemical properties [10].



In terms of modeling, computational methods have been developed to optimize the photocatalysis process using predictive modeling. This includes the use of artificial neural networks, Gaussian process regression, and neuro-fuzzy inference systems. This modeling is based on the input factors that affect the process such as concentration of the pollutant, irradiation time, pH level, and the dosage and structural feature of the catalyst [11]. This shows the importance of having an effective photocatalyst. Density functional theory (DFT) can provide information on electronic structures and reaction mechanisms which can be used to compute the band gap, valence and conduction bands, and other outcomes [12]. DFT is an essential tool to model TiO_2 , making it possible to understand how doping and surface functionalization modify the band gap, surface properties, and recombination rates [13]. This information can be used to optimize the photocatalytic process. Despite the wide range of benefits of DFT, there are limitations such as the structure and size of TiO_2 that under realistic environmental conditions lead to long simulation times [14]. The literature shows that the complexity of the photocatalytic process stems from intricate reactions, leading to models that simulate unrealistic scenarios. A survey of methods has been proposed such as machine learning; however, current developments in quantum algorithms pose a potentially positive outcome. This outcome arises due to the possibility of forming accurate electronic structure calculations that simulate chemical reactions under complex environmental conditions [15]. Previous studies have looked at the use of quantum phase estimation (QPE) for some molecules [16]. The cited study applies QPE to benzene and its chloro- and nitro-substituted derivatives to compute electronic ground states and the π - π^* excited singlet states. The method uses pseudo-natural orbitals as the basis for the wavefunction and calculations within the active space to identify the dominant electronic configurations for excited-state input preparation. The result of this study shows the ability to compute both ground and excited state energies within an accuracy margin, demonstrating the possibility of applying this method to other molecules that can help analyze more complex systems. Current NISQ hardware does not handle circuits that are too deep, especially for complex systems, while classical simulations have exponential growth with respect to qubit count. This shows that modeling larger molecules is infeasible without fault-tolerant quantum computing. Therefore, it is beneficial to estimate the resources of molecules for particular applications.

Given that there is no quantum resource estimation of titanium dioxide, this computational study provides an avenue towards a novel application for photocatalytic wastewater treatment. Pre-existing applications in the literature can provide comparisons with the current pipeline and be modified according to the application. For example, two studies focus on the design of corrosion-resistant materials and CO_2 utilization for green energy production [17]. Even though the application differs from the current study, the structure and pipeline are modeled similarly. Other benchmark comparisons are with the study of the active site of Mo-dependent nitrogenase, the iron molybdenum cofactor

(FeMoco), which can be used to assess the range of the results of the current study [18]. However, this validation method should not be considered one-to-one but should simply assess the range of our outcomes. The cited study predicts that quantum computers with around 100 logical qubits could solve the FeMoco mechanism within days to months. This computational study estimates the full resources of titanium dioxide for photocatalytic wastewater treatment. It is divided as follows: Section 2 discusses the methods and model assumptions, Section 3 provides the results and validation, Section 4 discusses the resource implications, Section 5 discusses the limitations and future work, and Section 6 describes the reproducibility infrastructure.

2 Methods and Model Assumptions

The target system is a minimal Ti_2O_4 cluster derived from the anatase polymorph of titanium dioxide. Anatase is the most photocatalytically active phase of TiO_2 due to its indirect band gap of approximately 3.2 eV [19]. The cluster geometry consists of two titanium atoms and four oxygen atoms with Ti–O bond lengths of approximately 1.93–1.97 Å. A finite cluster model is used because it admits a well-defined active space, and the cluster is treated as a closed-shell singlet with $S = 0$ and charge = 0. The atomic coordinates are given in Table 1. This is a simplification of the molecule, as this 6-atom cluster is a benchmark molecule used to test the resource estimates.

Table 1: Atomic Coordinates of the Ti_2O_4 Cluster

Atom	x (Å)	y (Å)	z (Å)
Ti	0.000	0.000	0.000
Ti	1.892	1.892	0.950
O	0.946	0.946	0.000
O	−0.946	−0.946	0.000
O	0.946	2.838	0.950
O	2.838	0.946	0.950

The initial stage involves the classical step to obtain the molecular orbital basis and extract the Hamiltonian integrals for the active space. A Hartree-Fock calculation was performed using PySCF to obtain the ground-state molecular orbitals [19]. An active space was selected because the full (76*e*, 118*o*) system is too large for direct simulation. This selection is based on the region dominated by the Ti 3d and O 2p levels. Table 2 shows the label, active space in terms of electrons and orbitals (*e*, *o*), and the corresponding number of spin orbitals. The smaller active space at (12*e*, 12*o*) is used to validate against classical full configuration interaction. The choice of the target is based on the correlation that occurs at the 3d and 2p levels, making it favorable given that DFT suffers from strong electron correlation. Initial DFT results are provided in Appendix A. By choosing the target and range of active spaces, the result can be extrapolated to slightly larger active spaces. Comparing the result to FeMoco serves as an initial benchmark to compare the ranges between the two systems, although both systems have distinct structures and energies.

Equation (4) shows the total electronic Hamiltonian within the active space, where p, q, r, s are indices of the active orbitals. The core energy includes the nuclear repulsion, the kinetic and nuclear attraction energies of the frozen electrons, and the Coulomb and exchange interactions, as shown in Equation (5). The effective one-electron integrals, depicted in Equation (6), show the potential of the core on the active electrons. The two-electron integrals within the active space are computed using the `ao2mo` module of PySCF.

Table 2: Active Space Selection from 8 to 48 Spin Orbitals

Label	Active Space	Spin-Orbitals
Small	(4e, 4o)	8
Medium	(8e, 8o)	16
Target	(12e, 12o)	24
Large	(16e, 16o)	32
XLarge	(20e, 20o)	40
XXLarge	(24e, 24o)	48

$$\hat{H} = E_{\text{core}} + \sum_{pq,\sigma} h_{pq}^{\text{eff}} \hat{a}_{p\sigma}^\dagger \hat{a}_{q\sigma} + \frac{1}{2} \sum_{pqrs} \sum_{\sigma\tau} (pq|rs) \hat{a}_{p\sigma}^\dagger \hat{a}_{r\tau}^\dagger \hat{a}_{s\tau} \hat{a}_{q\sigma} \quad (4)$$

$$E_{\text{core}} = E_{\text{nuc}} + \sum_{i \in \text{core}} 2h_{ii} + \sum_{i,j \in \text{core}} [2(ii|jj) - (ij|ji)] \quad (5)$$

$$h_{pq}^{\text{eff}} = h_{pq} + \sum_{i \in \text{core}} [2(pq|ii) - (pi|i q)] \quad (6)$$

Equation (4) is the second-quantized electronic Hamiltonian of the chosen active space. The indices p, q, r, s run over active spatial orbitals and σ, τ over spin. The first term E_{core} as denoted in Equation (5) is a constant with respect to the active electrons. It includes the nuclear-nuclear repulsion term, the kinetic energy plus electron-nucleus attraction summed over the doubly-occupied core orbitals, and the Coulomb $(ii|jj)$ minus exchange $(ij|ji)$ interaction among the frozen core electrons. Equation (6) contains the bare one-electron integral between active orbitals p and q plus the mean-field Coulomb and exchange potentials generated by the frozen-core electrons acting on the active electrons.

The Jordan-Wigner transformation maps the fermionic Hamiltonian to the qubit Hamiltonian, where each spin orbital is mapped to a single qubit [20]. Equation (7) shows the Hamiltonian as a linear combination of Pauli strings, where $\hat{P}_\ell$ is a tensor product of Pauli operators and α_ℓ are real coefficients. Equation (8) is the 1-norm of the Hamiltonian, the parameter that determines the depth of the quantum circuit.

$$\hat{H}_q = \sum_{\ell=1}^L \alpha_\ell \hat{P}_\ell, \quad \hat{P}_\ell \in \{I, X, Y, Z\}^{\otimes N} \quad (7)$$

$$\lambda = \sum_{\ell=1}^L |\alpha_\ell| \quad (8)$$

In Equation (7) the Hamiltonian takes the form of the linear-combination-of-unitaries (LCU) form through a sum of L Pauli strings. The number of QPE rounds required to resolve the energy to precision ε is shown in Equations (9)–(10) with ε set to 1.6 mHa. In the PauliLCU encoding, each Pauli term is treated as an independent entry in the linear combination of unitaries. The total T-gate count is the product of the T-gate cost per walk operator N_T^{walk} and the number of QPE rounds N_{QPE} .

$$N_{\text{QPE}} = \left\lceil \frac{\pi\lambda}{2\varepsilon} \right\rceil \quad (9)$$

$$N_T = N_{\text{QPE}} \cdot N_T^{\text{walk}} \quad (10)$$

A second representation is double factorization (DF), which decomposes the two-electron integral tensor and gives a much lower walk-operator cost than the PauliLCU encoding. In surface code error correction, logical qubits are mapped to physical qubits under a rotated surface code. In this study, the physical error rates are 10^{-3} , 10^{-4} , and 10^{-5} . Each logical qubit requires $2d^2$ physical qubits in the rotated surface code, where d is the code distance. The total physical qubit count is shown in Equation (11), and the wall-clock time assumes a cycle time of 1 μs and d code cycles per T-gate as seen in Equation (12).

$$N_{\text{phys}} = 2d^2 \cdot N_{\text{logical}} \quad (11)$$

$$T_{\text{wall}} = N_T \cdot d \cdot t_{\text{cycle}} \quad (12)$$

Tensor hypercontraction (THC) is a third encoding method introduced for chemistry resource estimation by Lee et al. [21]. This method decomposes the two-electron integrals as a contraction of an $N \times M$ matrix of interpolation vectors and an $M \times M$ core tensor, where M is the THC rank. Compared to the previous methods, THC takes advantage of a low-rank structure, leading to a smaller 1-norm and significantly fewer resources. Validation strategies are used with the first computing the FCI on the (12e, 12o) space, which can be extended to higher spaces. The correlation energy is compared with the QPE target precision of 1.6 mHa, which is chosen based on this molecule. The second strategy compares the T-gate scaling exponent against the growth of the Pauli terms under the Jordan-Wigner encoding. These outcomes are pre-computed using built-in libraries such as PyLIQTR and Google’s Qualtran. By installing and loading these libraries, it is possible to obtain and compare the resources required for anatase.

To simplify the scope of the problem, several assumptions are made. The physical model (anatase) is treated as a finite cluster with no periodic structure and fixed geometry, with atomic coordinates as in Table 1. The frozen-core approximation is held in active spaces where orbitals outside the active range are doubly occupied and not optimized, based on the active spaces in Table 2. Active spaces are chosen by HOMO/LUMO proximity around the Hartree-Fock Fermi level. The final outcomes are extrapolated to larger orbital spaces by assuming that the fitted power laws hold beyond the (24e, 24o) range studied directly.

3 Results and Validation

The initial classical calculations on the Ti_2O_4 cluster converged to -1994.6516 Ha for the Hartree-Fock method. The B3LYP-DFT calculation converged to -1998.4942 Ha. The HF HOMO-LUMO gap was 3.85 eV, while B3LYP gave a gap of 0.26 eV. The PauliLCU portion of Table 3 shows the PauliLCU resource estimates. The 1-norm λ ranged from 2.97 at (4e, 4o) to 184.6 at (24e, 24o) and scales sub-linearly because most Pauli coefficients are small in absolute value, with the dominant contributions coming from diagonal one-body terms. The total T-gate count increases from 3.6×10^7 at the smallest space to 2.7×10^{12} at the largest, while the logical qubit count grows only from 22 to 59. PauliLCU is the simplest encoding to implement, but the per-walk T-cost scales linearly in the

number of Pauli terms, which makes it the most expensive of the three encodings tested.

Table 3: PauliLCU, DF, and THC Estimates by Active Space

Active Space	Encoding	λ	QPE Bits	T-gates	Logical Qubits
(4e, 4o)	PauliLCU	2.97	12	3.60×10^7	22
	DF	7.86	14	1.70×10^7	29
	THC	6.71	14	1.74×10^7	34
(8e, 8o)	PauliLCU	15.26	15	3.91×10^9	33
	DF	34.85	16	4.71×10^8	42
	THC	28.46	16	1.63×10^8	50
(12e, 12o)	PauliLCU	37.43	16	4.10×10^{10}	41
	DF	79.62	17	3.02×10^9	54
	THC	66.02	17	5.61×10^8	65
(16e, 16o)	PauliLCU	69.64	17	2.58×10^{11}	47
	DF	142.34	18	1.29×10^{10}	63
	THC	112.44	18	1.49×10^9	78
(20e, 20o)	PauliLCU	117.90	18	1.29×10^{12}	54
	DF	229.15	19	4.87×10^{10}	74
	THC	181.41	18	1.86×10^9	91
(24e, 24o)	PauliLCU	184.65	18	2.69×10^{12}	59
	DF	339.60	19	8.21×10^{10}	83
	THC	268.59	19	5.07×10^9	105

The DF portion of Table 3 shows the DF results, which have a higher λ that is almost twice the PauliLCU value because the DF method absorbs two-body coefficients into a basis where individual coefficient magnitudes are larger. The result is a T-count that drops from 4.1×10^{10} to 3.0×10^9 at (12e, 12o) and from 2.7×10^{12} to 8.2×10^{10} at (24e, 24o), a reduction of approximately a factor of 30 at the largest active space. The logical qubit footprint grows by 24 qubits relative to PauliLCU due to the registers storing the DF eigenvalues and rotation angles. Despite these reductions, the gate counts remain large compared to those reported for FeMoco at much larger active spaces [21]. The THC estimates show that the 1-norm is reduced by a factor of approximately 1.2 to 1.3 relative to DF, as THC uses a low-rank structure in the two-electron tensor. The advantage of THC comes from the per-walk cost, leading to a much lower T-count at larger active spaces, with more than 100-fold reduction compared to PauliLCU and DF. Power-law fits give a scaling exponent of 3.1 for THC compared to 4.9 for DF and 6.3 for PauliLCU; the sample size is limited and should be extended further for a more accurate fit. This result agrees with the asymptotic forms predicted by Lee et al. [21] and shows the importance of using the appropriate encoding strategy. THC can be used when handling active spaces in the 50–100 orbital range, making it a practical method for chemically meaningful systems.

Table 4 combines the three encodings with the rotated surface-code overhead at a physical error rate of $p = 10^{-3}$, reporting the code distance d , total physical qubit count, and wall-clock time in days. The required code distance grows with system size as the total operation count grows, requiring stronger logical-error suppression. PauliLCU consistently demands the largest code distance because its T-count is the largest, while THC requires the smallest. The physical qubit counts are comparable across encodings at any given active space because the count is dominated by the number of logical qubits and distance d . However, the wall-clock time differs by orders of magnitude given that the number of T-

gates as observed in Table 3 is vastly different. At (24e, 24o), PauliLCU requires 842 days, DF requires 22 days, and THC requires 1.2 days. This disparity further emphasizes the importance of selecting a suitable encoding method. Despite the THC results being analytical estimates, they provide a practical avenue toward fault-tolerant quantum computing. THC at (24e, 24o) under $p = 10^{-3}$ runs in approximately 30 hours on approximately 119,000 physical qubits, within the 10^5 physical qubit budget projected for early fault-tolerant systems. The corresponding PauliLCU calculation, by contrast, would require operating that hardware for over two years continuously, which is not a practical experiment.

Table 4: Surface Code Overhead at $p = 10^{-3}$ Across Encodings

Active Space	Method	d	Physical Qubits	Wall (days)
(4e, 4o)	PauliLCU	17	30,056	0.01
	DF	17	34,102	0.00
	THC	17	36,992	0.00
(8e, 8o)	PauliLCU	21	55,566	0.95
	DF	19	51,984	0.10
	THC	19	57,760	0.04
(12e, 12o)	PauliLCU	23	75,118	10.91
	DF	21	74,088	0.73
	THC	19	68,590	0.12
(16e, 16o)	PauliLCU	25	96,250	74.74
	DF	23	98,394	3.43
	THC	21	95,256	0.36
(20e, 20o)	PauliLCU	27	122,472	403.93
	DF	23	110,032	12.95
	THC	21	106,722	0.45
(24e, 24o)	PauliLCU	27	129,762	841.60
	DF	23	119,554	21.87
	THC	21	119,070	1.23

Figure 1(a) shows three curves of T-gate count versus active orbitals. The blue squares are PauliLCU, the gold diamonds are DF, and the green circles are THC estimates. The dashed lines are power-law fits extrapolated to 100 orbitals. At every system size, THC requires fewer T-gates and fewer overall resources than DF or PauliLCU, and the gap widens at larger active spaces. PauliLCU scales as $N^{6.3}$ due to the number of Pauli terms, DF scales as $N^{4.8}$ because the per-walk cost is $\mathcal{O}(N\Xi)$ rather than $\mathcal{O}(L)$, and THC scales as $N^{3.1}$ because the per-walk cost is approximately $\mathcal{O}(N + M \log M)$. Figure 1(b) shows the classical-quantum crossover. The red triangles trace the classical FCI operation count, which grows combinatorially. At active spaces below 20 orbitals, the classical FCI cost is below the quantum cost, but at approximately 25 orbitals, the exponential classical curve crosses above both PauliLCU and DF, and THC crosses below the classical curve at approximately 20 orbitals. At 50 orbitals, the classical cost exceeds 10^{28} while the quantum cost is around 10^{14} , and the gap widens further at larger ranges.

Figure 2 shows the physical qubit count as a function of active orbitals at three physical error rates (10^{-3} , 10^{-4} , 10^{-5}) under the THC encoding. THC is chosen for the overhead analysis due to its favorable scaling for chemically meaningful active-space sizes [21]. The figure has three marker styles corresponding to the three error rates. At a fixed error rate, the physical qubit count grows with active

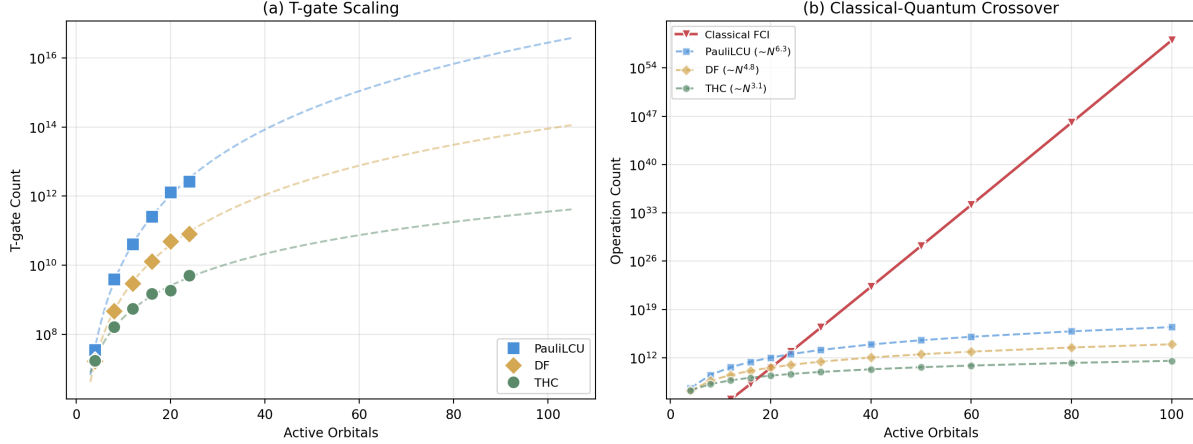


Figure 1: Resource estimates and classical-quantum behavior across active spaces from $(4e, 4o)$ to $(24e, 24o)$, with power-law fits extrapolated to 100 orbitals. (a) T-gate scaling for PauliLCU, DF, and THC. (b) Classical-quantum crossover comparing the three quantum encodings against the classical FCI operation count.

orbital count, but much more slowly than the T-gate count grows. The $(24e, 24o)$ THC calculation requires approximately 120,000 physical qubits at $p = 10^{-3}$ but only 13,000 at $p = 10^{-5}$. At fixed system size, the physical qubit count drops by roughly an order of magnitude as the physical error rate improves from 10^{-3} to 10^{-5} , because better physical qubits permit smaller code distances.

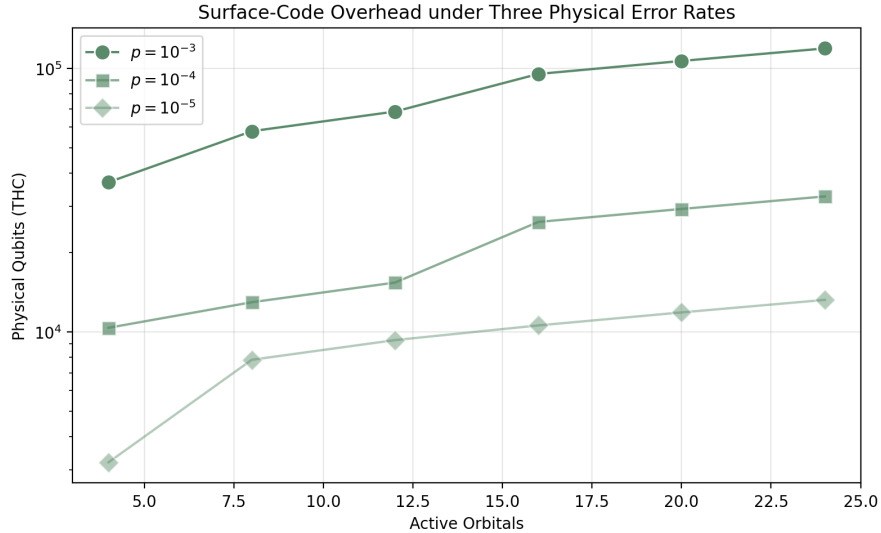


Figure 2: Physical qubit count versus active orbitals under THC encoding at three physical error rates: $p = 10^{-3}$, $p = 10^{-4}$, and $p = 10^{-5}$. Improving physical error rates reduces the surface-code distance and therefore the per-logical-qubit footprint.

To validate the results, full configuration interaction (FCI) was performed on the $(12e, 12o)$ target. This range was selected because FCI is computationally tractable here, although it would be prohibitive at larger active spaces. The validation verifies that the correlation energy missed by Hartree-Fock exceeds the QPE precision target. If the gap were smaller than the precision, chemical-accuracy QPE would not resolve any meaningful improvement over the mean-field reference. The Hartree-Fock energy in the active space converged to -26.416 Ha and the FCI energy to -26.455 Ha, giving a correlation energy of 39.69 mHa (1.08 eV). This is approximately 25 times the QPE precision target of 1.6 mHa (0.0435 eV), confirming that QPE at this precision provides information that cannot be captured by classical mean-field methods (HF, DFT). This outcome supports the validity of the

resource estimates as quantitative descriptions of the work required to obtain a chemically accurate ground-state energy on fault-tolerant hardware.

A second validation anchor is the comparison of fitted T-gate scaling exponents against asymptotic forms from the literature. The PauliLCU scaling of $N^{6.3}$ and the DF scaling of $N^{4.9}$ obtained from the six-point fit are within the range of values reported by Babbush et al. and Lee et al. for similar transition-metal Hamiltonians [21, 22], while the THC scaling of $N^{3.1}$ matches the asymptotic $\mathcal{O}(N^3)$ prediction from Lee et al. [21]. A third anchor is the comparison of THC estimates with the FeMoco benchmark. Lee et al. [21] report approximately 3.5×10^{10} T-gates, 2,196 logical qubits, and 4×10^6 physical qubits at $p = 10^{-3}$ for FeMoco at the (54e, 54o) active space. The extrapolation of our THC fit to the same active space gives 5.6×10^{10} T-gates and 2.5×10^5 physical qubits, a comparable range of T-gates and an order-of-magnitude lower physical-qubit footprint due to differences in λ and factory allocation between the two pipelines. It is important to note the differences between the two structures: this is not a one-to-one comparison but a feasibility benchmark that places TiO₂ resource requirements within the same regime as the most-studied transition-metal benchmark in the field, given that there is no prior study estimating quantum resources for photocatalysis.

4 Resource Implications

A realistic TiO₂ simulation for photocatalytic wastewater treatment would require 50–100 active orbitals to converge transition-metal correlation. The PauliLCU and DF analysis shows that all three encodings scale polynomially in this regime, so the practical question is the prefactor and scaling exponent rather than asymptotic tractability. At 50 active orbitals under PauliLCU encoding at $p = 10^{-3}$, the pipeline predicts approximately 3.5×10^{14} T-gates and 1.5×10^5 physical qubits, with a wall-clock time on the order of years. The classical FCI versus quantum QPE comparison shows the underlying separation: classical FCI scales combinatorially as $\left(\binom{N}{N/2}\right)^2$, while quantum QPE scales polynomially. At 25 orbitals the quantum cost first drops below the classical cost; at 50 orbitals the gap is approximately 14 orders of magnitude (10^{14} for QPE versus 10^{28} for FCI), and the gap widens further at higher orbital counts. At the (24e, 24o) cluster studied directly, PauliLCU requires 2.7×10^{12} T-gates and 842 days, while DF reduces the T-count to 8.2×10^{10} and the wall-clock time to 22 days. At the best-tested physical error rate of $p = 10^{-5}$, the (24e, 24o) PauliLCU calculation requires 54 logical qubits, 6.5×10^{10} T-gates, and approximately 280 days on 15,000 physical qubits. Extending PauliLCU to 25 orbitals would require 80–90 logical qubits and approximately 10^{12} T-gates, mapping to 15,000 physical qubits and 10^2 – 10^3 days of wall time at $p = 10^{-5}$; expanding further to 50 orbitals would require an order-of-magnitude increase in physical qubit count, approaching the projected 10^5 physical qubit budgets of next-decade fault-tolerant systems.

Beyond PauliLCU and DF, tensor hypercontraction provides a substantial further reduction that changes the practical outlook. At the (24e, 24o) target, THC requires approximately 5×10^9 T-gates and 105 logical qubits, three orders of magnitude below the corresponding PauliLCU estimate, with a wall-clock time of approximately 30 hours at $p = 10^{-3}$ and under 10 hours at $p = 10^{-5}$. Extrapolating THC to a 50-orbital active space gives approximately 4×10^{10} T-gates and 2.4×10^5 physical qubits at $p = 10^{-3}$, falling within the resource envelope reported by Lee et al. [21] for the FeMoco active site at (54e, 54o). This places converged TiO₂ photocatalysis simulations in the same difficulty class as the most-studied transition-metal benchmark in the field, rather than in a regime requiring qualitatively new algorithmic developments. The dominant lever is no longer the encoding choice but the physical error rate of the underlying hardware: improving from $p = 10^{-3}$ to $p = 10^{-5}$ reduces the physical qubit footprint by an order of magnitude and the wall-clock time by a factor of three, while changing the encoding from PauliLCU to THC reduces the T-count by three orders of magnitude. Together,

these levers are multiplicative, and the headline finding is that for the same physical hardware, the choice of encoding determines whether the calculation takes hours, days, or years.

5 Limitations and Future Work

The principal limitation is the assumption of substantial overlap between the Hartree-Fock initial state and the true ground state, which depends on the molecule chosen. For transition-metal oxides at large active spaces, this overlap can be small, in which case the QPE success probability decreases as the square of the overlap and the effective T-count scales as the reciprocal of this probability. This dependency is system-specific and active-space-specific: the $(12e, 12o)$ target was verified to have substantial Hartree-Fock content via the FCI overlap implied by the 39.69 mHa correlation gap, but extrapolation to 50–100 orbitals may encounter regimes where this assumption fails and where the resource estimates inflate by factors of two to ten. Initial-state preparation via matrix-product-state initialization or ADAPT-VQE warm start would close this gap but is not currently included in the cost model. Second, the wall-clock estimates assume ideal parallelism between the data block and the magic-state distillation factories; realistic factory throughput may slow the calculation by a factor of two to three or require additional physical qubits to maintain the assumed pace. Third, the physical model does not include all aspects of the photocatalytic system, such as the relationship between the photocatalyst and the substrate. The resources reported here represent the cost of the algorithmic kernel rather than the complete cost of the photocatalytic process, including the relationship with pollutants and their degradation.

A separate set of challenges sits below the algorithmic layer and concerns whether the assumed fault-tolerant hardware stack can actually be built at the scale these estimates require. The surface code overhead formulas assume a physical error rate at or below 10^{-3} , which has only been demonstrated on small qubit registers. Scaling from tens of physical qubits to the 10^5 – 10^6 qubit range required for chemically meaningful TiO_2 active spaces introduces engineering challenges in cryogenic control wiring, classical decoder latency, and crosstalk suppression that are not captured by the per-cycle error model. A second hardware challenge is the rate of magic-state production: the cost formulas assume that a small number of distillation factories can supply T-states at the rate the QPE walk operator consumes them, but at the T-counts reported here, the consumption rate is on the order of 10^5 T-states per second, which exceeds the throughput of any factory architecture currently demonstrated in hardware. A third challenge concerns control-system bandwidth: each surface-code cycle requires synchronized measurement and feedback across the entire logical qubit array, and the $1\ \mu\text{s}$ cycle time imposes strict latency requirements on the classical decoder that become increasingly difficult to meet as the code distance grows. These challenges occur in different directions in terms of hardware, algorithms, and applications.

Future work will address these challenges by integrating matrix-product-state initialization into the pipeline and improving the geometric model to include passivated clusters and periodic anatase slabs with selected active spaces. These extensions can more accurately model the resource estimates, which is beneficial for projecting fault-tolerant hardware needs.

6 Reproducibility

The complete pipeline is publicly available at https://github.com/HishamKazim/PHY765_Project_Kazim, with the tagged release v1.0 corresponding to the version that produced the tables and figures in this paper. The repository is organized as seven Jupyter notebooks executed in order, covering DFT analysis, integral generation, the three encodings, FCI validation, and surface-code overhead analysis. Parameters such as geometry, basis set, active-space sweep, QPE precision, and surface-code

error rates are collected in a single `config.yaml` file, and a wrapper script `regenerate_figures.py` executes the full pipeline end-to-end. The code requires Python 3.11 and the dependencies listed in `requirements.txt`.

References

- [1] S. Khan, M. Naushad, M. Govarthanan, J. Iqbal, and S. M. Alfadul, “Emerging contaminants of high concern for the environment: Current trends and future research,” *Environ. Res.*, vol. 207, p. 112609, May 2022, doi: 10.1016/j.envres.2021.112609.
- [2] A. M. Briggs et al., “Musculoskeletal Health Conditions Represent a Global Threat to Healthy Aging: A Report for the 2015 World Health Organization World Report on Ageing and Health,” *Gerontologist*, vol. 56, no. Suppl 2, pp. S243–S255, Apr. 2016, doi: 10.1093/geront/gnw002.
- [3] N. Bolong, A. F. Ismail, M. R. Salim, and T. Matsuura, “A review of the effects of emerging contaminants in wastewater and options for their removal,” *Desalination*, vol. 239, no. 1–3, pp. 229–246, Apr. 2009, doi: 10.1016/j.desal.2008.03.020.
- [4] A. Al-Bsoul et al., “Optimal conditions for olive mill wastewater treatment using ultrasound and advanced oxidation processes,” *Sci. Total Environ.*, vol. 700, p. 134576, Jan. 2020, doi: 10.1016/j.scitotenv.2019.134576.
- [5] N. Shams Jalbani et al., “Synthesis of new functionalized Calix[4]arene modified silica resin for the adsorption of metal ions: Equilibrium, thermodynamic and kinetic modeling studies,” *J. Mol. Liq.*, vol. 339, p. 116741, Oct. 2021, doi: 10.1016/j.molliq.2021.116741.
- [6] A. Egea-Corbacho Lopera, S. Gutiérrez Ruiz, and J. M. Quiroga Alonso, “Removal of emerging contaminants from wastewater using reverse osmosis for its subsequent reuse: Pilot plant,” *J. Water Process Eng.*, vol. 29, p. 100800, Jun. 2019, doi: 10.1016/j.jwpe.2019.100800.
- [7] H. Premkumar, “Synthesis, characterization and photocatalytic performance of polyvinyl alcohol/zinc oxide nanocomposites: A comprehensive study,” *Nano-Structures & Nano-Objects*, vol. 39, p. 101208, Sep. 2024, doi: 10.1016/j.nanoso.2024.101208.
- [8] X. Gao and X. Meng, “Photocatalysis for Heavy Metal Treatment: A Review,” *Processes*, vol. 9, no. 10, p. 1729, Sep. 2021, doi: 10.3390/pr9101729.
- [9] J. Jiang et al., “In Situ Ti^{3+} /N-Codoped Three-Dimensional (3D) Urchinlike Black TiO_2 Architectures as Efficient Visible-Light-Driven Photocatalysts,” *Ind. Eng. Chem. Res.*, vol. 56, no. 28, pp. 7948–7956, Jul. 2017, doi: 10.1021/acs.iecr.7b01693.
- [10] X. Wang, S. Liu, S. Lin, K. Qi, Y. Yan, and Y. Ma, “Visible Light Motivated the Photocatalytic Degradation of P-Nitrophenol by Ca^{2+} -Doped AgInS_2 ,” *Molecules*, vol. 29, no. 2, p. 361, Jan. 2024, doi: 10.3390/molecules29020361.
- [11] É. Nascimbén Santos, Z. László, C. Hodúr, G. Arthanareeswaran, and G. Veréb, “Photocatalytic membrane filtration and its advantages over conventional approaches in the treatment of oily wastewater: A review,” *Asia-Pacific J. Chem. Eng.*, vol. 15, no. 5, pp. 1–29, 2020, doi: 10.1002/apj.2533.
- [12] D. D. K. Wayo, L. Goliatt, and M. D. Ganji, “DFT and hybrid classical-quantum machine learning integration for photocatalyst discovery and hydrogen production,” *Reviews in Chemical Engineering*, vol. 41, no. 8, pp. 741–774, Sep. 2025, doi: 10.1515/revce-2025-0022.
- [13] S. Ricardo, R. Augusto, and L. Henrique, “Quantum Chemistry Applied to Photocatalysis with TiO_2 ,” *InTech eBooks*, Jul. 2017, doi: 10.5772/intechopen.69054.
- [14] D. Selli, G. Fazio, and C. Di Valentin, “Using Density Functional Theory to Model Realistic

- TiO₂ Nanoparticles, Their Photoactivation and Interaction with Water,” *Catalysts*, vol. 7, no. 12, p. 357, Nov. 2017, doi: 10.3390/catal7120357.
- [15] S. Gusarov, “Advances in Computational Methods for Modeling Photocatalytic Reactions: A Review of Recent Developments,” *Materials*, vol. 17, no. 9, p. 2119, Apr. 2024, doi: 10.3390/ma17092119.
- [16] Y. Ino, M. Yonekawa, H. Yuzawa, Y. Minato, and K. Sugisaki, “Workflow for practical quantum chemical calculations with a quantum phase estimation algorithm: electronic ground and π - π^* excited states of benzene and its derivatives,” *Physical Chemistry Chemical Physics*, Jan. 2024, doi: 10.1039/d4cp03454f.
- [17] N. Nguyen et al., “Quantum computing for corrosion simulation: workflow and resource analysis,” *npj Quantum Information*, vol. 12, no. 1, Jan. 2026, doi: 10.1038/s41534-025-01171-1.
- [18] M. Reiher, N. Wiebe, K. M. Svore, D. Wecker, and M. Troyer, “Elucidating reaction mechanisms on quantum computers,” *Proceedings of the National Academy of Sciences*, vol. 114, no. 29, pp. 7555–7560, Jul. 2017, doi: 10.1073/pnas.1619152114.
- [19] Q. Sun et al., “PySCF: the Python-based simulations of chemistry framework,” *WIREs Comput. Mol. Sci.*, 2018.
- [20] P. Jordan and E. Wigner, “Über das Paulische Äquivalenzverbot,” *Z. Phys.*, 1928.
- [21] J. Lee et al., “Even more efficient quantum computations of chemistry through tensor hypercontraction,” *PRX Quantum*, vol. 2, p. 030305, 2021.
- [22] R. Babbush, D. W. Berry, J. R. McClean, and H. Neven, “Quantum simulation of chemistry with sublinear scaling in basis size,” *npj Quantum Information*, vol. 5, no. 1, Nov. 2019, doi: 10.1038/s41534-019-0199-y.

A DFT Analysis

For the DFT analysis, the PySCF package was used to perform both Hartree-Fock (HF) and DFT calculations to obtain the electronic structure and orbital energies. The active space of (12e, 12o) was used to target the Ti 3d orbital and O 2p orbital. Table 5 shows the key results in terms of total energy, the HOMO-LUMO gap, the experimental gap, the active space, and the 1-norm λ of the Hamiltonian.

Table 5: Results of DFT and HF Analysis

Property	HF	B3LYP (Hybrid DFT)
Total Energy (Hartree)	−1994.65	−1998.49
HOMO-LUMO Gap (eV)	3.85	0.26
Experimental Gap, bulk anatase (eV)		3.2
Active Space	12 electrons, 12 spatial orbitals (24 spin-orbitals)	
Hamiltonian 1-norm λ		339.20

The total energy is the total electronic energy of the cluster. HF gives the mean-field result, where each electron experiences an averaged potential from the other electrons but neglects electron correlations. B3LYP is a hybrid DFT approach that combines HF exchange with correlation, and gives a lower energy by approximately 3.84 Ha due to the inclusion of correlation. The HOMO-LUMO gap shows two opposite errors: HF overestimates the gap (3.85 eV) compared to the experimental value of 3.2 eV, while B3LYP severely underestimates it at 0.26 eV. The B3LYP result is a known artifact of the bare-cluster geometry: the small Ti₂O₄ fragment has unsaturated dangling bonds at every

oxygen, and the resulting near-frontier orbital structure is dominated by surface states rather than bulk-like band edges. Neither value is meant to reproduce a converged bulk DFT or experimental result; the cluster is a benchmark molecule whose role is to test the resource-estimation pipeline. To further illustrate this behavior, Figure 3(a) shows the frontier orbital energies for the B3LYP method, with large energy shifts occurring between the H-1 and H-2 levels, which justifies the choice of active space. Figure 3(b) compares the band gap across the HF, hybrid DFT, and experimental approaches, showing the disparities discussed above.

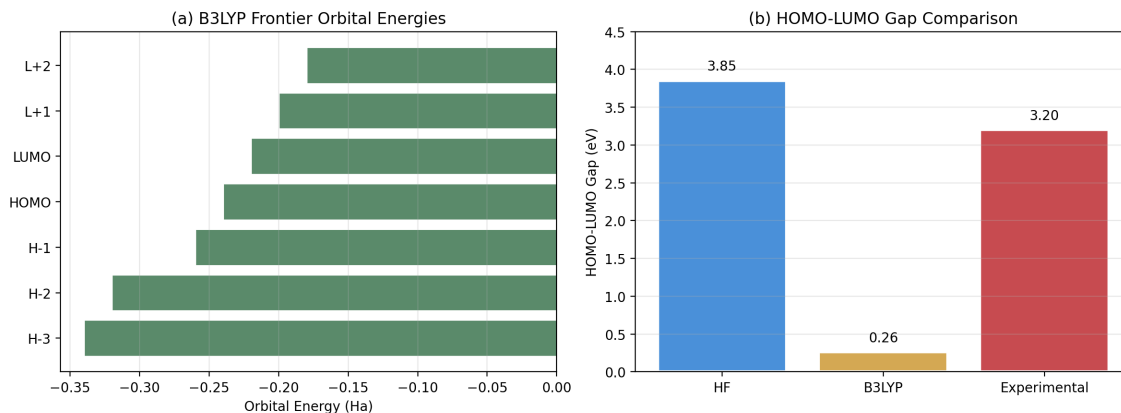


Figure 3: Cluster analysis using DFT: (a) Frontier orbital energies and (b) HOMO-LUMO gap comparison across HF, B3LYP, and experimental values.